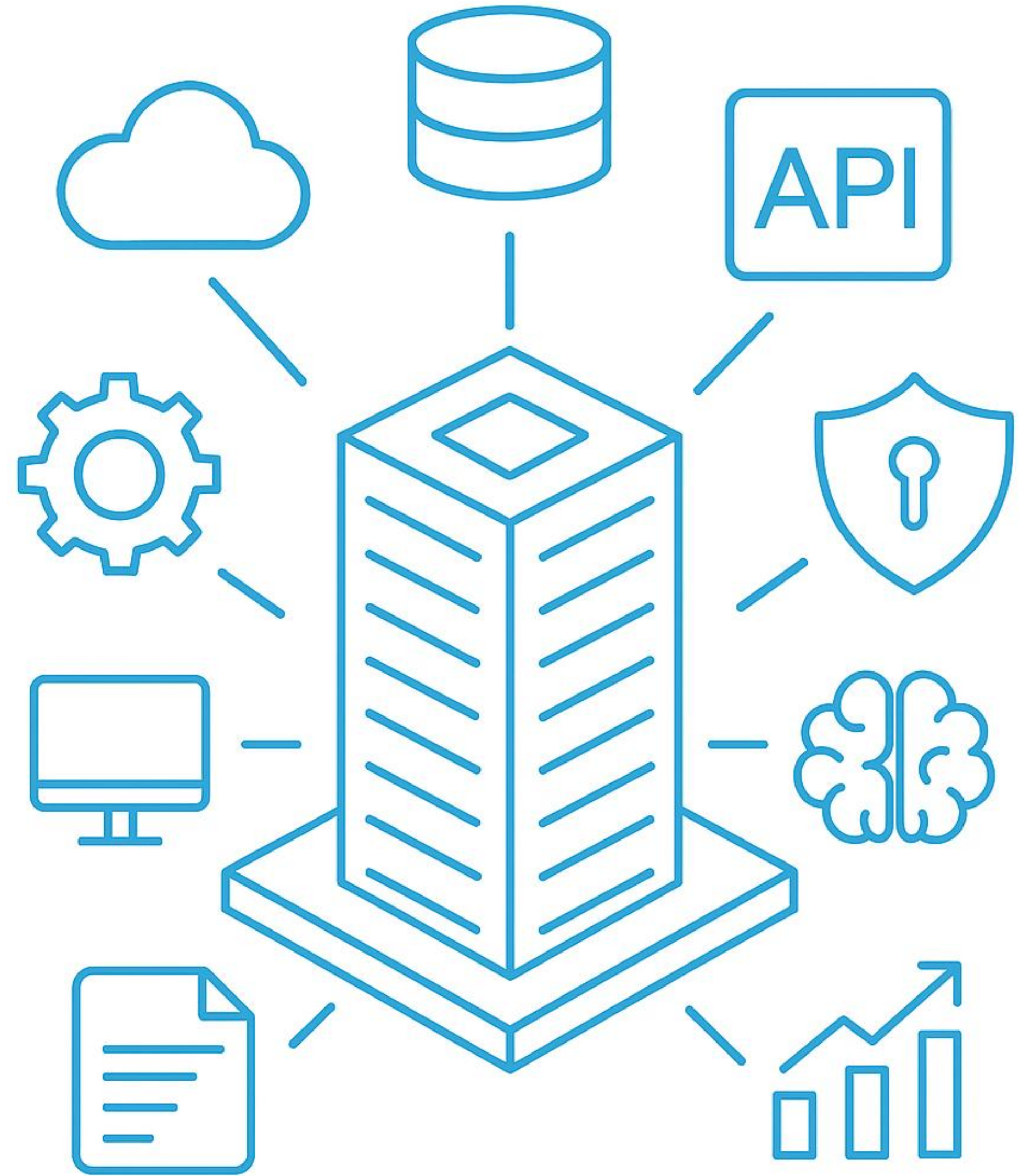
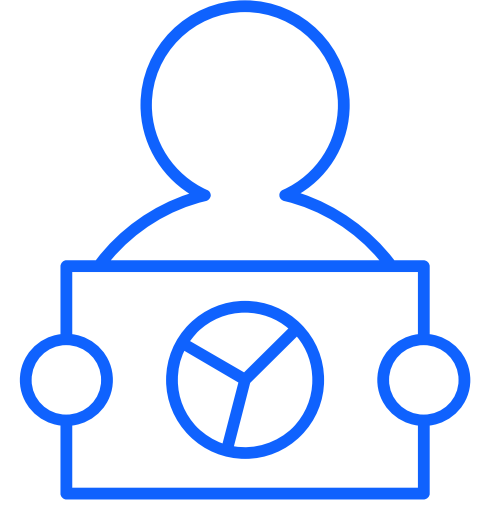
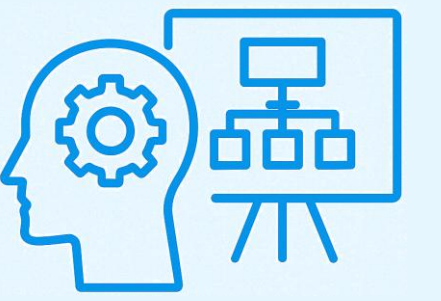


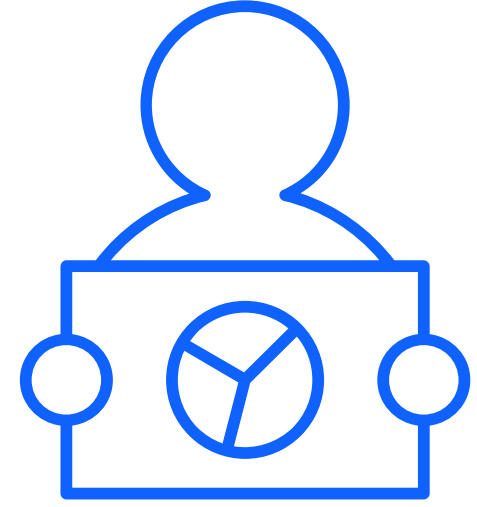
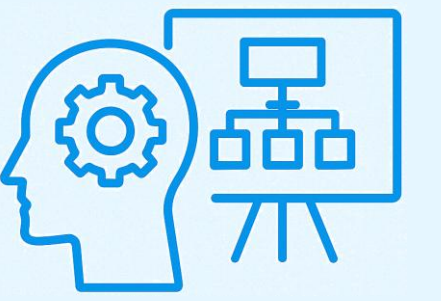
Validation and Viability

Version 7

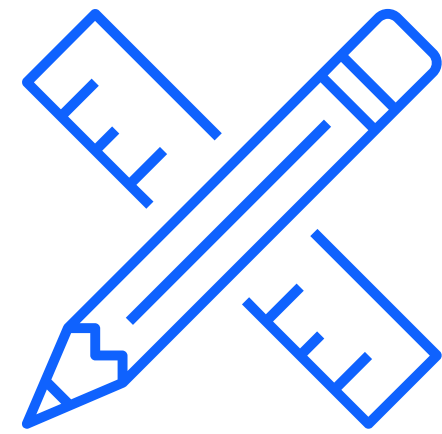
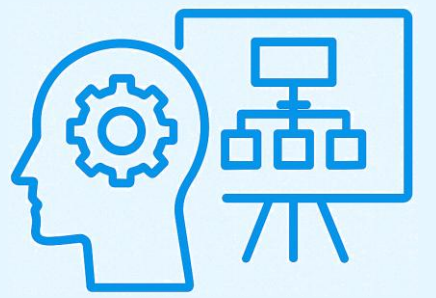




- Introducing and objectives
- Viability
- Validation
- Approaches to viability and validation
- Summary and references



- **Introducing and objectives**
- ↳ Viability
- ↳ Validation
- ↳ Approaches to viability and validation
- ↳ Summary and references



At the end of this module, you should be able to:

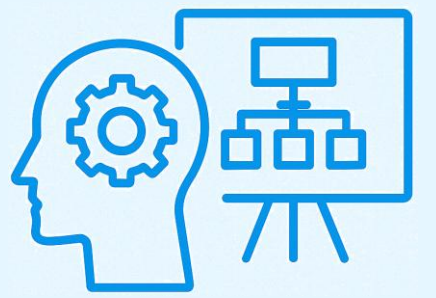
- **Describe**

- Technical reviews and their relationship to the quality assurance and testing processes
- How to identify resources available to help with viability assessments and technical reviews
- How to incorporate full life-cycle validation and verification into architectural thinking

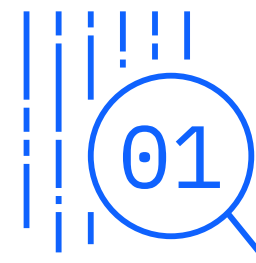
- **Identify**

- Those who are concerned about the architecture's quality: the Architect's target audiences and stakeholders
- How to determine the validity and viability of an architecture
- How to deal with risk: removal, containment, and acceptance

Why are Viability and Validation important in Architectural Thinking?

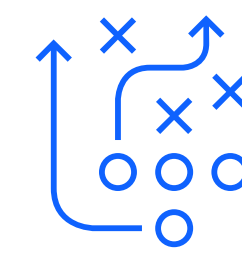


Humans are imperfect and so are solutions created by them. Fixing flawed solutions at an early stage of the project is much cheaper than at a later stage.

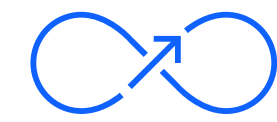


Reviews and assessments are used to ensure that the architecture (and its documentation) is ready, ensuring that it:

- can be implemented and delivered – *Viability*
- satisfies the functional requirements and qualities – *Validation*

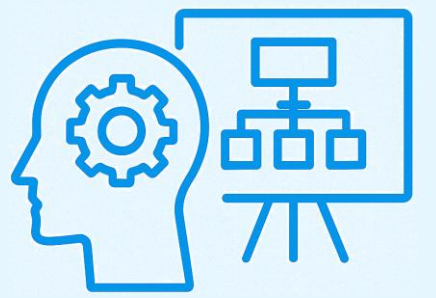


The Unified Method Framework work product for documenting the architecture assessment is Viability Assessment.



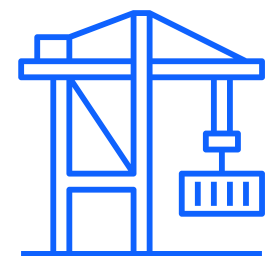
The validation and viability of the architecture are an important element of the overall approach to testing across the whole solution life cycle.

Common causes of troubled projects



Project Design

- Fixed price for combined phases
- Vague or misunderstood completion criteria
- Customer responsibilities not clearly identified and documented
- Assumptions not clearly identified and documented
- Concessions during negotiations
- Pricing concessions, “low-balling”
- Poorly written proposals or SOWs
- Inaccurate project estimates
- Not performing QA reviews
- Lack of transition, marketing to delivery



Project Delivery

- Project skills deficiencies
- Lack of, or inadequate, project management
- Not performing QA reviews and action plans
- Lack of line management oversight and support
- Not implementing or exercising change control
- Starting a phase before completing the preceding phase
- Unplanned turnover of key project team members



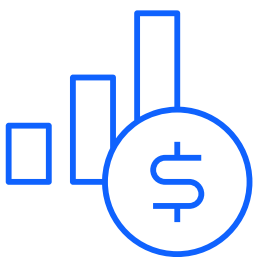
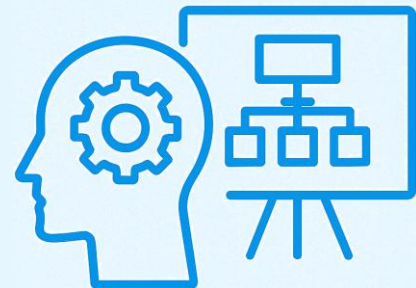
Summary

The common causes for troubled projects are known, and almost all are avoidable.

Remember:

- Viability assessments and other forms of validation and QA can help prevent your project from becoming troubled.
- The lists and assessments cannot help unless the recommended actions are taken up. As an Architect, take the lead.

Even most well thought solution are imperfect



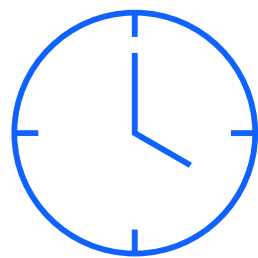
Cost
Budgets are limited. Even if a project starts off well-funded, new cost constraints may cause solution architectures to change radically from initial design.



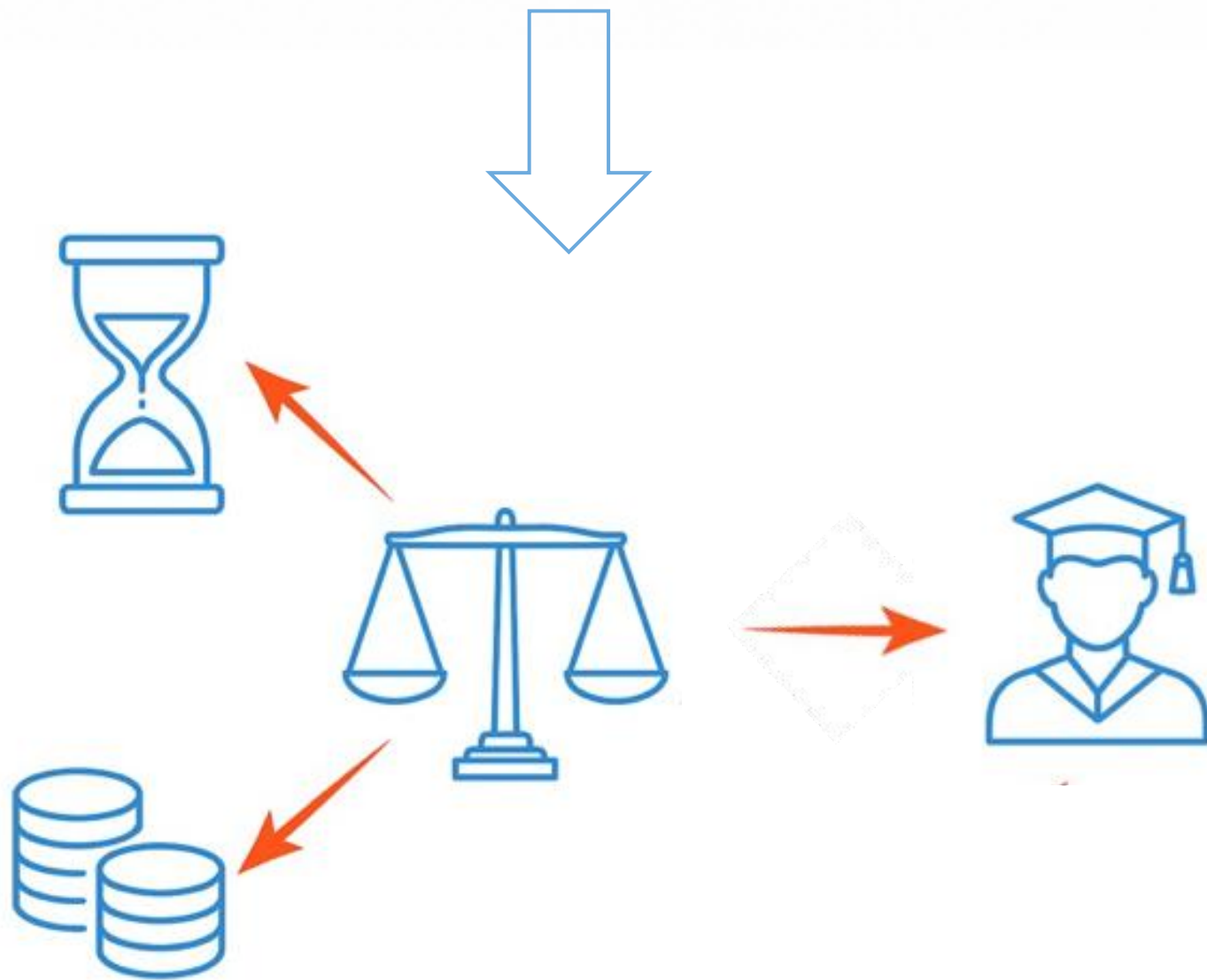
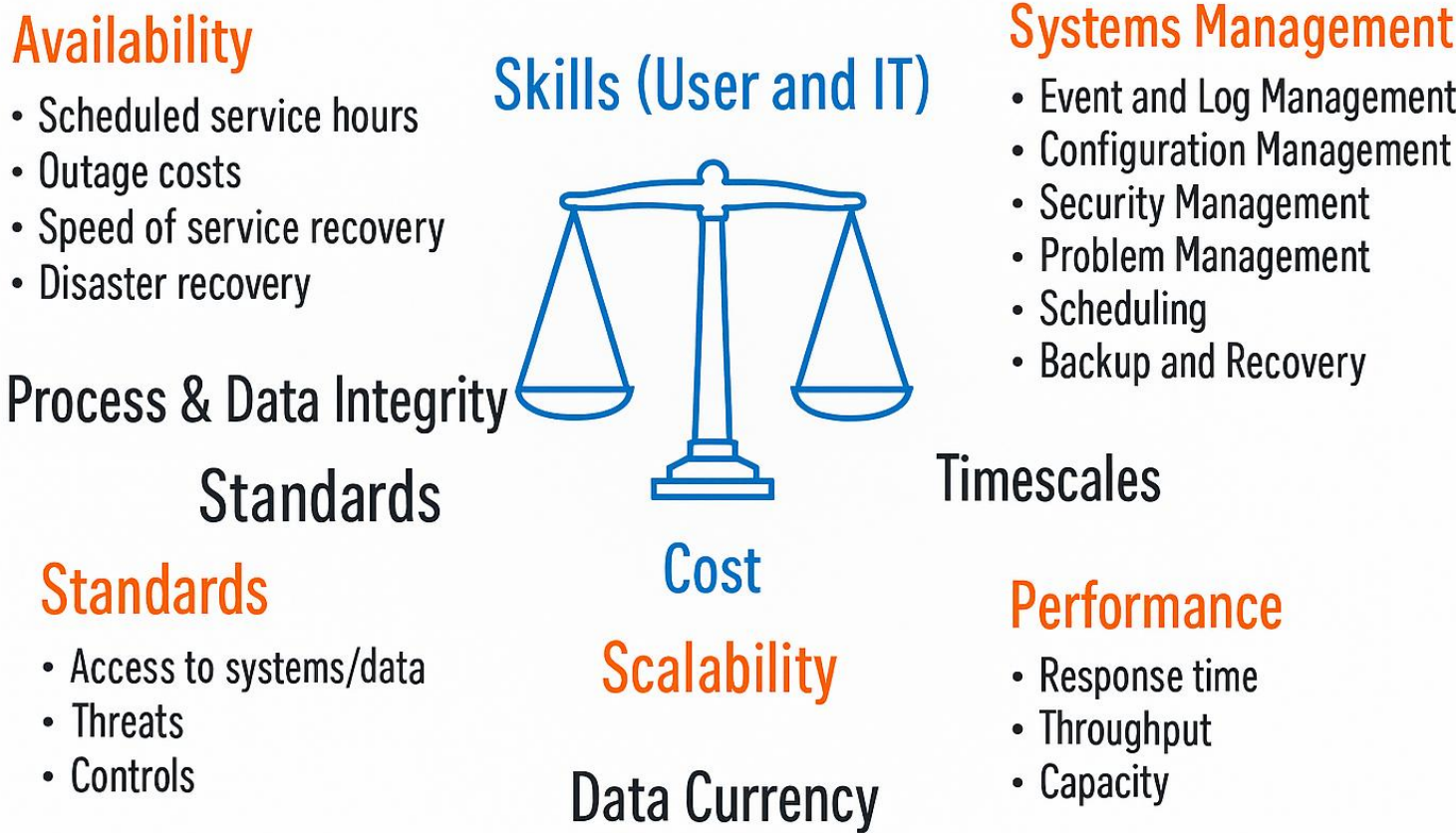
Information
“As you know more, you know more”. New information: requirements, legislation, competition causes change to the business. The architecture must change to accommodate new information constantly.

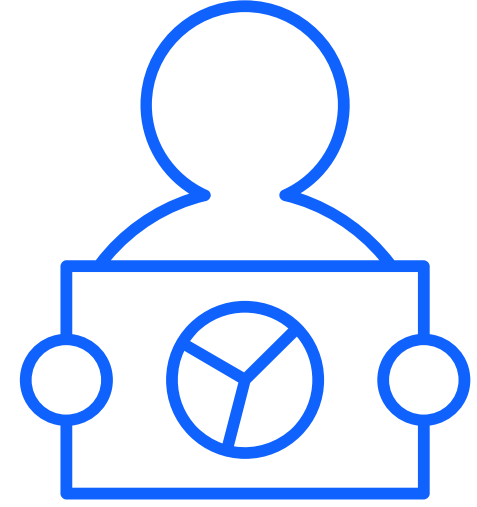
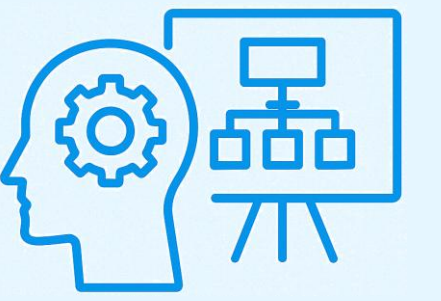


Skilled resources
Throughout a long-lived project, staff comes and goes.

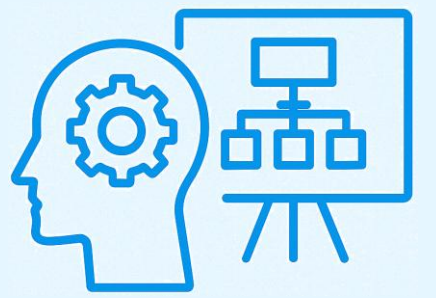


Time
Architects never have sufficient time, particularly at the beginning of the project when they have the most influence.

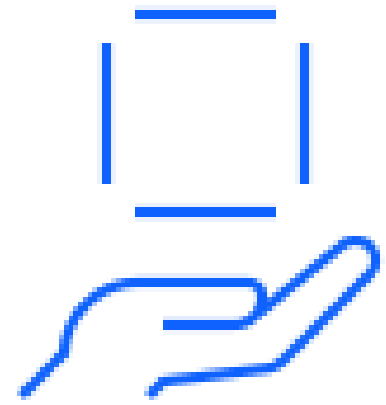




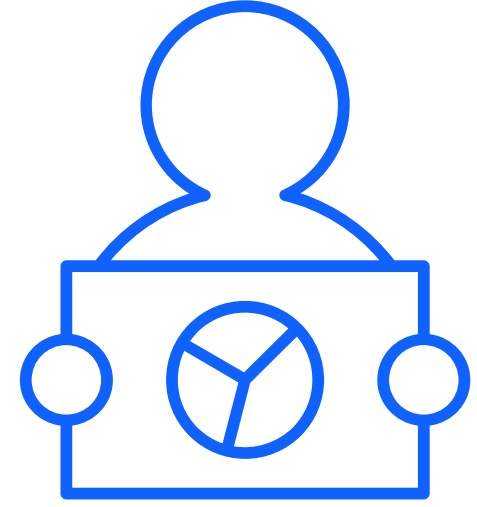
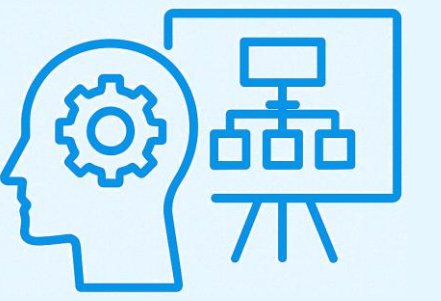
- ↗ Introducing and objectives
- **Viability**
- ↘ Validation
- ↘ Approaches to viability and validation
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Solution Architects performing the Viability Assessment¹:

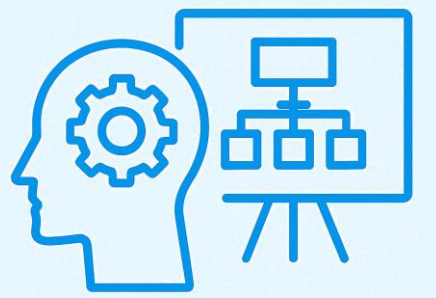


- **Assess the architecture.** Does it fit the wider context? Does it satisfy the wider needs of the enterprise:
 - As-is operational environment
 - To-be enterprise architecture
- **Confirm that the deployed solution will be operable and maintainable with operations**
- **Understand the context of the architecture. What are the key requirements and commitments?**
 - Assess whether the proposed solution is possible:
 - Can it be designed and developed within the available means?
 - Will it work? Will it perform? Will it be available?
- **Determine the project team's ability to deliver.**
 - Are the right resources in place to ensure success?
 - Is the necessary infrastructure in place?
- **Identify and assess technical risks.**
 - What risks must be raised and formally managed? What are their probable impacts? How can these be contained?

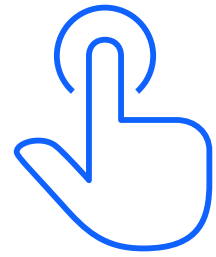


- Introducing and objectives
- Viability
- **Validation**
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What is validation?



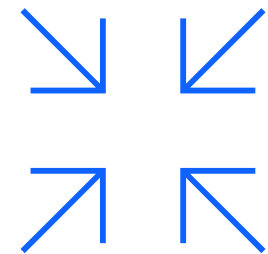
Considerations:



When examining an architecture's validity, consider the following:

- Does it do what is required of it? Does it meet the sponsor's needs?
- Does it do it well enough? Is it over engineered, or is it vulnerable in some way?

Acceptance criteria

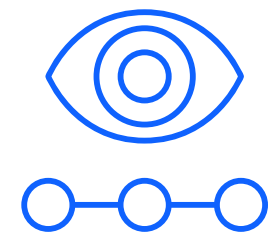


Solution Architects ensure that the proposed solution is acceptable to the various stakeholders:

- The key criterion is that it satisfies the Functional and Nonfunctional requirements
- This is achieved through the use of Requirements Traceability and Verification Matrix (RTVM). RTVM is used throughout the whole project, not just at the end

RTVM

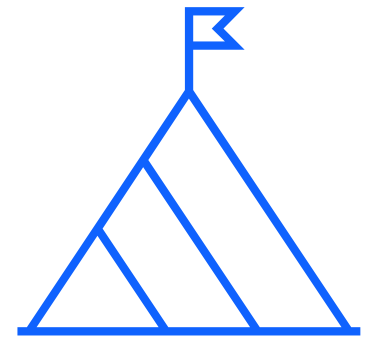
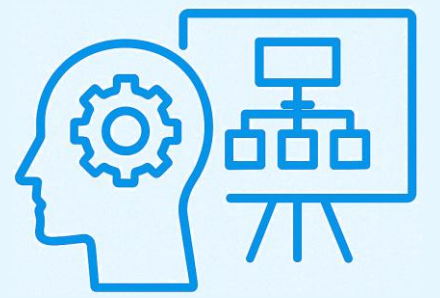
Requirements Traceability and Verification Matrix



- Ensures completeness of architecture and design through mapping of the requirements (both FRs and NFRs) to the solution components
- Provides the basis for test planning and execution
- Provides a framework for risk management and mitigation
- Provides the basis of clear and timely client communication with regard to schedule, risk, and financial impacts

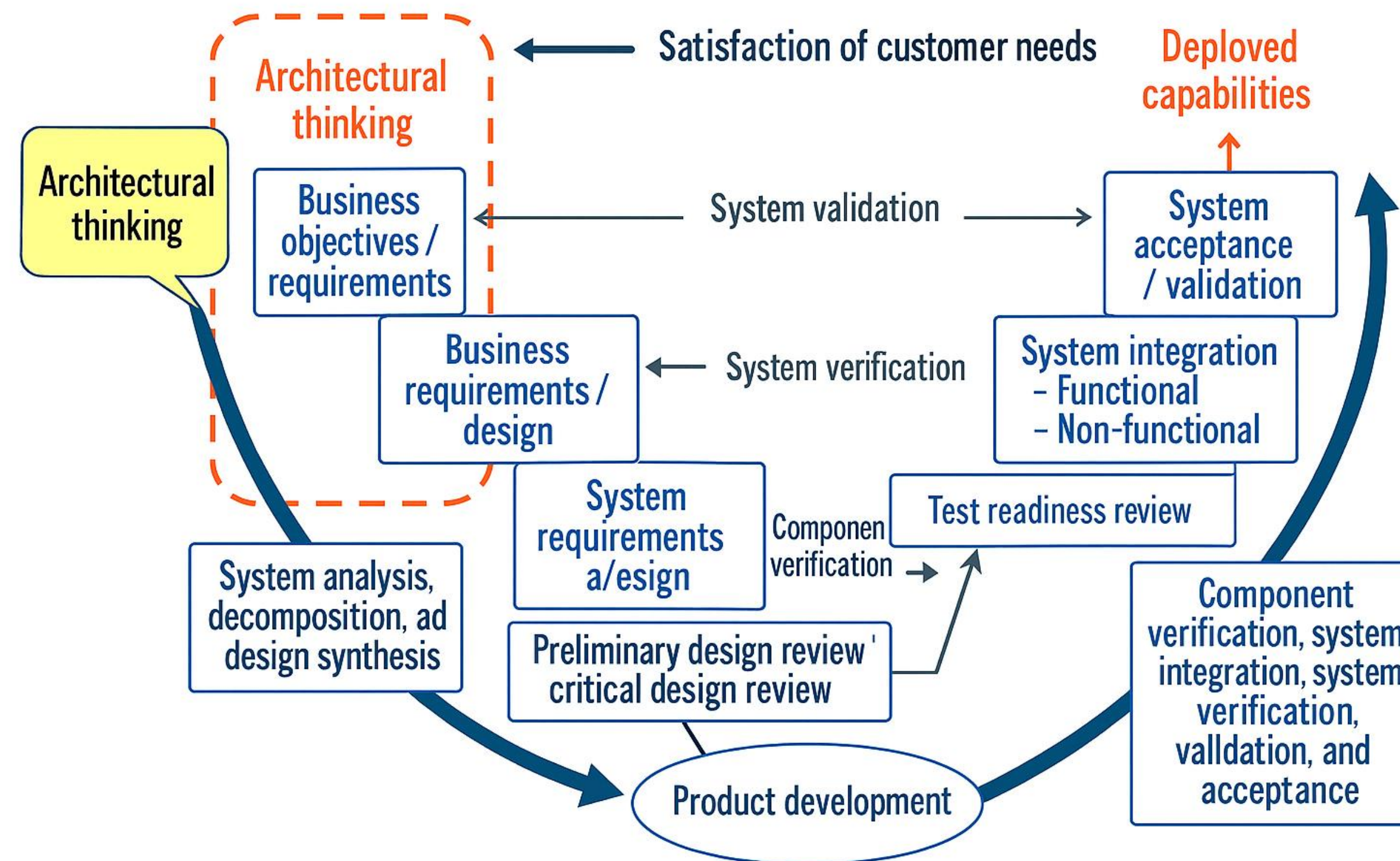


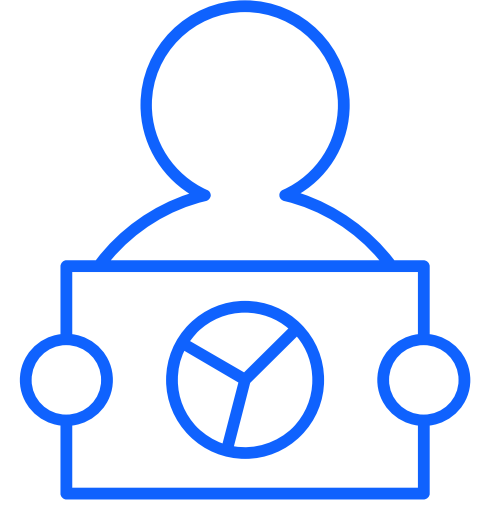
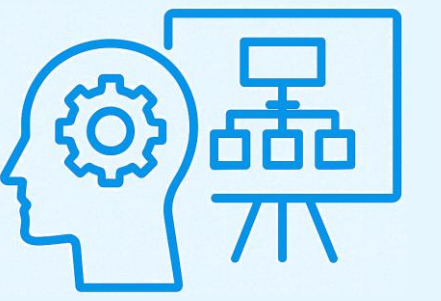
Validation is not an after the fact concern; it is addressed throughout the project



The testing V model and system engineering baseline reviews are a good starting point for planning validation and verification activities.

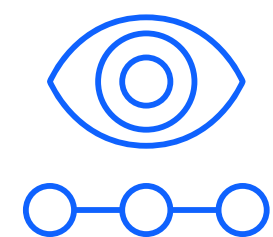
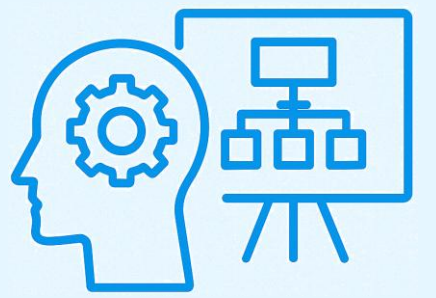
The V model ensures that all tests relate to requirements, and all requirements have an associated test case.





- ↗ Introducing and objectives
- ↗ Viability
- ↗ Validation
- **Approaches to viability and validation**
- ↘ Summary and references

IBM manages overall project viability using CVM



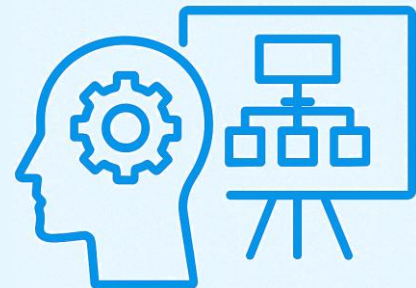
TDA and ITR (for complex, cross-LOB, and/or cross-IOT deals) are quality checkpoints¹

- Validate that the integrated end-to-end proposed solution will fulfill client requirements, is technically viable, and can be delivered by IBM
- Validate that the integrated project estimates and schedules being proposed are complete and reasonable
- Identify and assess the technical risks, with containment plans in place to mitigate the technical risks
- Led by an Architect within the cross-LOB Solution Design Architect community

¹These are not mandatory. Some business units mandate them if the engagement size exceeds certain clip level.

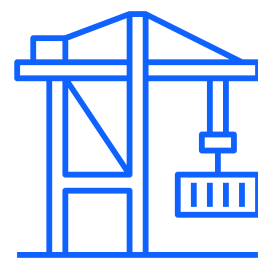


An approach to risk assessment: IBM GSRisk tool



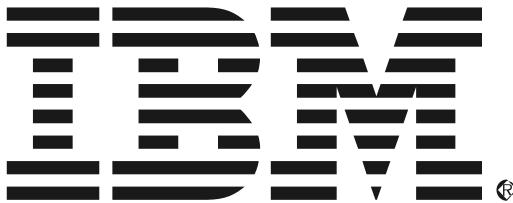
IBM GS Risk is a risk management tool that is available for all IBMers through w3. Originally developed for IBM Consulting engagements, GS Risk asks a large number of questions relating to the technical and delivery elements of the proposed solution:

- **Team Skills:** Has the team used the proposed toolset before?
- **Requirements:** Did IBM develop the requirements or are we relying on the customer or a third party to develop them?
- **Technologies:** Are the proposed technologies mature? Are we relying on third-party hardware or software for all or part of the solution?



GS Risk then assigns a risk assessment for the proposal that is used as input to the pricing process

- **Risk assessment** is one of the key inputs to pricing considerations. High risk attracts a higher price category.
- Don't be afraid of medium to high risk. IBM Consulting projects come under the medium or higher risk categories when first assessed.



Application Management Services

Transition/Transformation

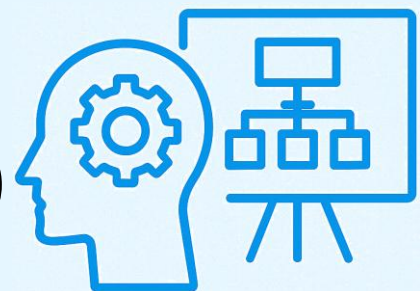
F.5. Cost Case

Check ALL that apply.

- ☒ Transition cost are not aligned with current bid metrics.
- ☐ Transformation cost are not aligned with current bid metrics.
- ☐ Cost for infrastructure, connectivity or environmental support during T&T are not sufficiently included.
- ☐ HR cost for rebadging staff is not sufficiently included.
- ☐ There are no risks or problems in this area.

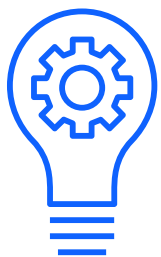
Sample question in a GS Risk assessment for an AMS project

Assessments should also address assumptions, issues, risks and dependencies: RAID



Risk

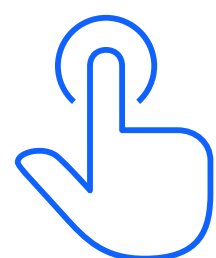
A possibility of something (usually negative) happening within the duration of a project



Assumption

Things that the Solution Architect has taken on faith or otherwise have no direct reason to be thought of as true. Often, assumptions are implicit but are not recognized as such

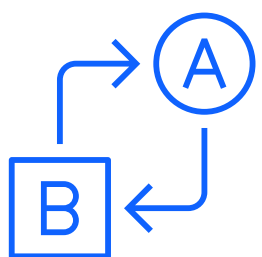
- Assumptions have to be verified and then become requirements or constraints. If found to be incorrect, the solution needs to be re-assessed.



Issue

Something that is causing or will cause the solution to deviate from the requirements baseline

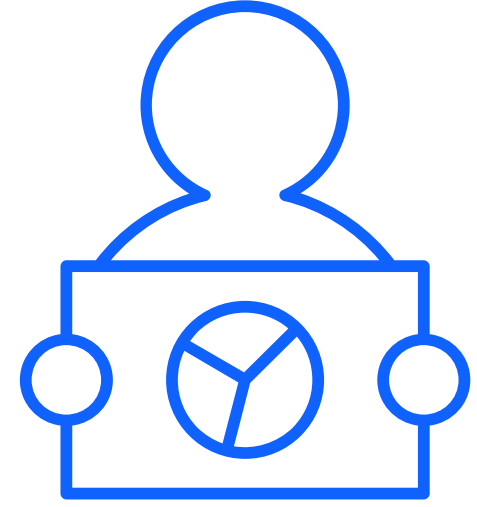
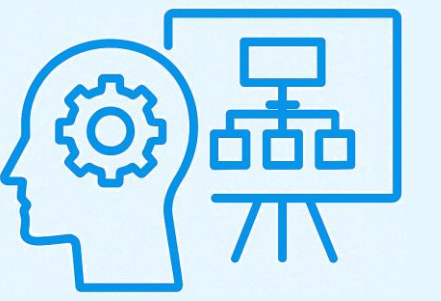
- Issues have to be addressed, and recommendations for their resolution have to be established, agreed, and acted upon.



Dependency

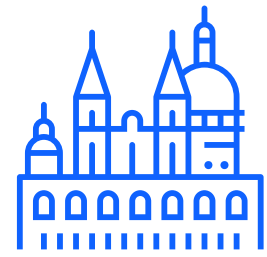
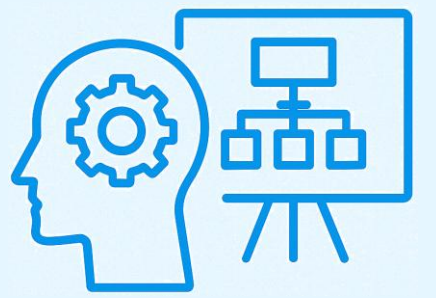
Something that a project expects to be in place or resolved before the project completes. Dependencies may be:

- Business-centric: “The market will be ready for this project.” “The legislator will clarify that this approach to data sharing is legal.”
- IT-centric: “IBM will fix this bug.” “We’ll find a way of synchronizing these databases.”
- Solution-centric: “The security server has the capacity to take our extra load.” “The warehouse management system will be ready in time.”

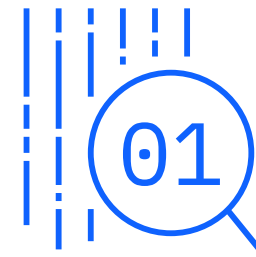


- Introducing and objectives
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- ➔ **Summary and references**

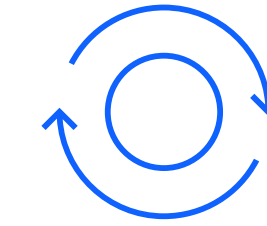
Summary



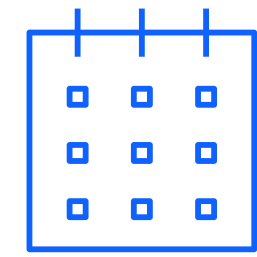
A “good” architecture must be valid – i.e. solution to satisfy the requirements, and be viable – can be delivered and will work.



Architectural assessment includes asking questions to challenge the architecture and associated delivery approach. A working architectural process delivers the base for a viable solution and its assessment.



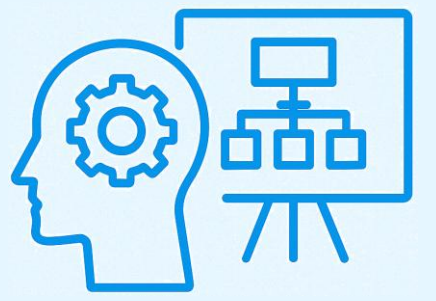
Validity and viability must be assessed throughout the project, as well as before (presales), and after (operations).



Assessments can follow a well-known approach: Issues, Risks, Dependencies and assumptions:

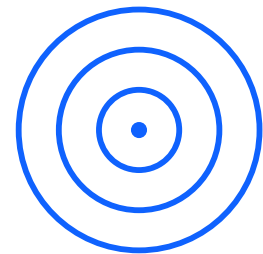
- Issues need to be resolved, or architecture needs to be re-assessed.
- Risks that are of high probability and/or high impact must be removed, contained, or accepted.
- Dependencies need to be documented and managed during solution delivery.

References for viability and risk assessment



IBM Unified Method Framework: <https://yourlearning.ibm.com/activity/PLAN-9740404306B5>

- Viability Assessment guidelines from method
- RTVM guidelines from method
- Solution Assurance: <https://yourlearning.ibm.com/activity/LT-4616>



Quality Assurance

- IBM Global Services Worldwide Quality Assurance Course https://yourlearning.ibm.com/activity/ALM-COURSE_3954973



Risk assessment

- GS Risk tool: <https://w3.ibm.com/#/apps/gsrisk>
- Managing Risk: Methods for Software Development, Hall E., Addison-Wesley, 1998, ISBN 0-201-25592-8



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